Chahat Raj

PhD Candidate in Computer Science | George Mason University

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Research Interests

My research focuses on NLP, particularly the evaluation and alignment of large language models. I study socio-cultural biases and other harms in multilingual and multimodal systems and develop methods to mitigate them, advancing trustworthy and responsible AI. My work has been published at ACL, EMNLP, AIES, AAAI, and other venues.

Education

May 2027 | **George Mason University** — **Fairfax, VA**
Aug 2022 | Ph.D. in Computer Science
Advisors: **Ziwei Zhu**, **Antonios Anastasopoulos**

Dec 2025 | **George Mason University** — **Fairfax, VA**
Aug 2022 | Master's in Computer Science

Aug 2021 | **Delhi Technological University** — **Delhi, India**
Aug 2019 | Master's in Information Systems (Thesis)
Advisor: **Priyanka Meel**

May 2019 | **Indira Gandhi Delhi Technical University for Women** — **Delhi, India**
Aug 2015 | Bachelor's in Computer Science & Engineering

Experience

Aug 2026 | **Amazon** — **Seattle, WA**
May 2026 | Applied Scientist Intern | Mentor: **Ren Pang**, Manager: **Zheng Qi**
› *Developed a parameter-efficient safety-steering technique for Amazon Bedrock Guardrails foundation models.*

Aug 2025 | **George Mason University, University of Washington** — **Fairfax, VA**
May 2024 | Graduate Research Assistant | Advisors: **Ziwei Zhu**, **Antonios Anastasopoulos**, **Aylin Caliskan**
› *Societal simulations, model unlearning, and cognitive alignment for bias mitigation.*

Sep 2024 | **University of Washington | Tech Policy Lab, The Information School** — **Seattle, WA**
May 2024 | Visiting Scholar | Advisor: **Aylin Caliskan**
› *Bias, cognition, and mitigation in vision-language models.*

Aug 2023 | **George Mason University | School of Computer Science** — **Fairfax, VA**
May 2023 | Graduate Research Assistant | Advisor: **Ziwei Zhu**
› *Fairness, interpretability, and robustness in transformer-based fake news detection.*

Jul 2022 | **University of Technology Sydney | School of Computer Science** — **Sydney, AU**
Jun 2021 | Visiting Scholar | Advisor: **Mukesh Prasad**
› *Cyberbullying, AI for Aboriginals, modeling disaster fundraising, and ML for clinical diagnosis.*

Selected Publications

full list here

[C.12] **VIGNETTE: Socially Grounded Bias Evaluation for Vision-Language Models**
Chahat Raj, Bowen Wei, Aylin Caliskan, Antonios Anastasopoulos, Ziwei Zhu
ACL 2026 (Main) **Oral, SAC Highlight Award**

[C.11] **Talent or Luck? Evaluating Attribution Bias in Large Language Models**
Chahat Raj, Mahika Banerjee, Jinhao Pan, Aylin Caliskan, Antonios Anastasopoulos, Ziwei Zhu
ACL 2026 (Findings)

[C.10] **Knowing Bias, Doing Better: Mitigating Social Bias in LLMs via Know-Bias Neuron Enhancement**
Jinhao Pan, Chahat Raj, Anjishnu Mukherjee, Sina Mansouri, Bowen Wei, Shloka Yada, Ziwei Zhu
ICML 2026

[C.9] **Measuring South Asian Biases in Large Language Models**
Mamnuya Rinki, Chahat Raj, Anjishnu Mukherjee, Ziwei Zhu
TrustNLP @ ACL 2026

[C.8] **Discovering Bias Associations through Open-Ended LLM Generations**
Jinhao Pan, Chahat Raj, Ziwei Zhu
AAAI 2026

[C.7] **What's Not Said Still Hurts: A Description-Based Evaluation Framework for Measuring Social Bias in LLMs**
Jinhao Pan, **Chahat Raj**, Ziyu Yao, Ziwei Zhu
EMNLP 2025 (Findings)

[C.6] **Toward Inclusive Language Models: Sparsity-Driven Calibration for Systematic and Interpretable Mitigation of Social Biases in LLMs**
Prommy Sultana Hossain, **Chahat Raj**, Ziwei Zhu, Jessica Lin, Emanuela Marasco
EMNLP 2025 (Findings)

[C.5] **BiasDora: Exploring Hidden Biased Associations in Vision-Language Models**
Chahat Raj, Anjishnu Mukherjee, Aylin Caliskan, Antonios Anastasopoulos, Ziwei Zhu
EMNLP 2024 (Findings)

[C.4] **Breaking Bias, Building Bridges: Evaluation and Mitigation of Social Biases in LLMs via Contact Hypothesis**
Chahat Raj, Anjishnu Mukherjee, Aylin Caliskan, Antonios Anastasopoulos, Ziwei Zhu
AIES 2024 **Oral**

[C.3] **Global Voices, Local Biases: Socio-cultural Prejudices across Languages**
Chahat Raj*, Anjishnu Mukherjee*, Ziwei Zhu, Antonios Anastasopoulos
EMNLP 2023 (Main)

[C.2] **SALSA: Salience-Based Switching Attack for Adversarial Perturbations in Fake News Detection Models**
Chahat Raj*, Anjishnu Mukherjee*, Hemant Purohit, Antonios Anastasopoulos, Ziwei Zhu
ECIR 2024

[C.1] **True and Fair: Robust and Unbiased Fake News Detection via Interpretable Machine Learning**
Chahat Raj, Anjishnu Mukherjee, Ziwei Zhu
AIES 2023

Under Review

[S.3] **Bias-Sensitive but Not Bias-Specific: Supervision-Free Fairness Intervention in Vision-Language Models**
Prommy Sultana Hossain, **Chahat Raj**, Ziwei Zhu, Jessica Lin
In Submission

[S.2] **Debias It Yourself: Teaching LLMs Cognitive Bias-Mitigation Interventions**
Chahat Raj, Sina Mansouri, Aylin Caliskan, Antonios Anastasopoulos, Ziwei Zhu
In Submission

[S.1] **Inoculation Prompting: Implicit Bias Mitigation in Large Language Models**
Yongxu Sun, **Chahat Raj**, Aylin Caliskan
In Submission

Selected Projects

- **AI, Power & Society** — Auditing Diffusion Models for Generational and Representational Errors [Report]
- **Computer Vision** — Cultural Image Translation for Human Figures [Report]
- **AI: Ethics, Policy & Society** — Implicit Association Test in Multimodal Generative Models [Report]

Teaching Assistantship

- **CS 478 Natural Language Processing, GMU** — *Instr.: Antonios Anastasopoulos* — Fall '23
- **CS 584 Theory/Application Data Mining, GMU** — *Instr.: Ziwei Zhu* — Fall '25
- **CS 484 Theory/Application Data Mining, GMU** — *Instr.: Keren Zhou* — Spring '26
- **COMP 502 Mathematical Foundations of Computing, GMU** — *Instr.: William D Ellis* — Fall '23
- **CS 112 Introduction to Python Programming, GMU** — *Instr.: Shahnaz Kamberi* — Fall '22, Spring '23
- **CS 108 Introduction to Computer Programming, GMU** — *Instr.: Sara Lynn Gopalakrishna, Bella Yin* — Spring '24

Student Mentorship

- **Daksha Rajagopal** [TJHSST → Purdue CS] — Modeling normative ethics and cross-cultural moral reasoning
- **Angela Liu** [TJHSST → Bryn Mawr College] — Assessing moral attributional biases in ambiguous contexts in LLMs
- **Mahika Banerjee** [TJHSST → KU Leuven] — Measuring attribution bias, coauthor on the ACL Findings '26 paper
- **Mamnuya Rinki** [MS Thesis, GMU → IntraFi] — Mitigating South Asian Biases, coauthor on TrustNLP @ ACL '26 paper
- **Yongxu Sun** [MS, UW → PhD, UW] — Inoculation Prompting by in-context learning to debias LLM decisions
- **Diwita Banerjee** [MS, GMU → DoorDash] — Assessing cuisine biases; scientific figure hallucination detection
- **Hasini Jasthi** [TJHSST] — Accent and dialect disparities in voice assistants for Global South
- **Sina Mansouri** [MS, GMU] — Long Context Memory Information Retrieval in LLM Agents
- **Sharanya Garika** [CS 678 Advanced NLP] — Identifying intersectional bias in LLMs across Indo-Aryan languages
- **Aksh Patel** [CS 678 Advanced NLP] — Identifying intersectional bias in LLMs across Indo-Aryan languages
- **Samruddhi Deshmukh** [CS 795 AI Ethics] — Auditing Text-to-Image Generation Models for Social Representation
- **Abhi Krishna** [CS 795 AI Ethics] — Auditing Text-to-Image Generation Models for Social Representation

Awards & Grants

Outstanding Oral Presentation Award, '26	*First place, SoC PhD Research Symposium*	GMU
Digital Innovation Impact Award, '26	*SoC PhD Research Symposium*	GMU
Third Place Winner, '26	*Graduate research poster competition*	GMU
SAC Highlight Award, '26	*VIGNETTE; top 1.5% of accepted papers*	ACL
Outstanding Academic Award, '26	*Outstanding Academic Achievement Award*	GMU
Outstanding GTA Award, '26	*Outstanding Graduate Teaching Assistant Award*	GMU
Outstanding Reviewer Award, '26	*Reviewer recognition*	EACL
CAHMP Fellowship, '24	*Social biases in question answering*	GMU
Best Paper Award, '24	*A Psychological View to Social Bias in LLMs*	MASC-SLL
Summer Graduate Research Award, '23	*Explainability and robustness of debiasing*	GMU
Researcher Access Award, '25	*USD 1,000 in API credits*	OpenAI
Travel Grant, '24 & '23	*USD 2,000; San Jose and Montreal*	AAAI/ACM AIES
Research Excellence Award, '23	*Commendable Research Award; INR 50,000 each*	DTU
Research Excellence Award, '22	*Commendable Research Award; INR 50,000 each*	DTU
Graduate Scholarship, '19	*INR 297,000 from AICTE for qualifying GATE*	DTU

Talks

PhD Student Panel	*Invited Panelist, CS 700*	George Mason, VA
Debias It Yourself	*Selected Talk, SoC Research Symposium '26*	George Mason, VA
Rethinking Social Bias in NLP	*Invited Talk, MilaNLP, Fall '26*	Bocconi, Remote
Ethics and Bias in NLP	*Guest Lecture, CS 678 Advanced NLP, Fall '26*	George Mason, VA
VIGNETTE	*Talk at ACL '26*	San Diego, CA
LLM Ethics: Bias and Mitigation	*Guest Lecture, CS 678 Advanced NLP, Fall '24*	George Mason, VA
LLM Ethics: Bias and Mitigation	*Guest Lecture, CS 782 Advanced ML, Spring '24*	George Mason, VA
Breaking Bias, Building Bridges	*Talk at AAAI/ACM AIES '24*	San Jose, CA
Breaking Bias, Building Bridges	*Talk at SouthNLP '24*	Emory, GA
Breaking Bias, Building Bridges	*Talk at MASC-SLL '24*	Johns Hopkins, MD
Research Seminar	*Invited Talk, CS Department, Summer '24*	George Mason, VA
Ethics in NLP	*Guest Lecture, CS 478 NLP, Fall '23*	George Mason, VA
Text Tokenization and Probability	*Guest Lecture, CS 478 NLP, Fall '23*	George Mason, VA
True and Fair	*Talk at AAAI/ACM AIES '23*	Montreal, QC
Cyberbully Monitoring	*ML4AU Research Showcase '21*	Remote

Program Committee

Conference Reviewer	ARR '23-'26, ICLR '26-'27, NeurIPS '23, '26, CHI '26, AAAI AIA '26, COLM '25-'26, IASEAI '26-'27, AIES '23-'26, CSCW '24, ACM TIST
Workshop Reviewer	TrustNLP '24-'26, LTEDI @ EACL '24, EACL Demo '26-'27, ACL Demo '26, NAACL Demo '26-'27, DravidianLangTech @ ACL, LT-EDI @ ACL
Ethics Reviewer	ACL Rolling Review
Sub-Reviewer	NAACL '25, ACL '24, EMNLP '24, AIES '24

References

Dr. Ziwei Zhu	*Assistant Professor, George Mason University*
Dr. Antonios Anastasopoulos	*Associate Professor, George Mason University*
Dr. Aylin Caliskan	*Associate Professor, University of Washington*